

MACROECONOMIC RESEARCH

ECONOMIC ANALYSIS REPORT

Economic Cycle Framework

Five-Dimension Assessment, Market Performance, and Late-Cycle Pathways

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A five-dimension framework for assessing growth, employment, inflation, financial conditions, credit and late-cycle pathways.

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Economic Cycle Framework: Five-Dimension Assessment, Market Performance, and Late-Cycle Pathways

Part I: Five-Dimension Economic Cycle Summary Table

Growth	Employment	Inflation	Financial Conditions	Credit	Cycle Assessment
Rising	Improving	Low	Accommodative	Repairing	Early Recovery
Stable	Healthy	Near 2%	Neutral	Stable	Goldilocks Expansion
Accelerating	Strong	Rising	Beginning to Tighten	Still Loose	Reflation
Very Strong	Extremely Tight	High	Continuously Tightening	High Leverage	Overheating
Positive but Slowing	Low Hiring, Low Layoffs	Falling but Still Elevated	Restrictive	Tail-End Stress	Late Cycle / Soft Landing
Slowing or Near Zero	Beginning to Deteriorate	Rising or Elevated	Policy Dilemma	Increasing Stress	Stagflation
Negative Growth	Rising Layoffs	Falling	Beginning to Ease	Spreads Widening	Recession / Hard Landing
Deep Negative Growth	Large-Scale Unemployment	Falling Rapidly	Emergency Easing	Credit Freeze	Credit Crisis

I. Five-Dimension Economic Cycle Summary Table

Part II: Growth Indicators and Growth-State Assessment

1. Core Growth Indicators

The growth dimension should focus on the following four indicators:

- 1. **Real GDP, quarter-over-quarter annualized**
- 2. **Real Private Domestic Final Purchases**
- 3. **Real GDI and the average of GDP and GDI**
- 4. **Atlanta Fed GDPNow**

Real GDP measures the change in total economic output, but it can be distorted by inventories, government spending, and net exports.

Real Private Domestic Final Purchases mainly include private consumption and business fixed investment. It is often a better measure of underlying private-sector demand.

Real GDI measures economic activity from the income side. GDP measures activity from the expenditure side. In theory, GDP and GDI should be equal, but statistical discrepancies frequently arise. The average of GDP and GDI can reduce noise from either measure.

Atlanta Fed GDPNow provides a real-time estimate of current-quarter real GDP growth. It is most useful for determining whether the current-quarter growth outlook is being revised upward or downward.

2. Growth-State Thresholds

Growth State	Real GDP, Annualized	Real Private Domestic Final Purchases	GDPNow and GDI Signals	Typical Cycle
Rising	Recovering from negative or near-zero growth to above 1%	Recovering from negative or very weak growth to above 1%	GDPNow revised upward repeatedly; GDI stops declining	Early Recovery
Stable	Approximately 1.5%–2.5%	Approximately 1.5%–2.5%	GDPNow revisions remain limited; GDP and GDI move in the same direction	Goldilocks Expansion

Growth State	Real GDP, Annualized	Real Private Domestic Final Purchases	GDPNow and GDI Signals	Typical Cycle
Accelerating	Improves by at least 0.5–1.0 percentage point, often reaching 2.5%–3% or more	Rises above 2%–3%	GDPNow revised upward repeatedly; consumption and investment improve together	Reflation
Very Strong	Usually above 3%	Usually above 3%	GDPNow materially exceeds potential growth; GDP and GDI are both strong	Overheating
Positive but Slowing	Still positive, usually around 1%–2%, but clearly below previous levels	Still positive, but falls toward 1%–2%	GDPNow revised modestly downward; GDI weaker than GDP	Late Cycle / Soft Landing
Slowing or Near Zero	Approximately 0%–1%	Near zero or below 1%	GDPNow repeatedly revised downward; business investment and real consumption weaken	Stagflation or Pre-Recession
Negative Growth	Below 0%	Below 0% or near zero for several periods	GDPNow turns negative; GDP and GDI weaken together	Recession / Hard Landing
Deep Negative Growth	Usually below negative 2%	Clearly negative	GDPNow falls sharply; consumption, investment, and production contract broadly	Credit Crisis

3. Detailed Growth-State Combinations

Rising

“Rising” growth does not simply mean GDP is positive. It means the economy is recovering from contraction or stagnation.

Typical signals include:

- Real GDP turns positive after contraction;
- Private final demand rises from near zero to above 1%;
- GDPNow is revised upward several times;

- PMI new orders begin to recover;
- Real consumption and residential investment improve;
- The gap between GDP and GDI narrows.

This combination typically corresponds to an early recovery.

Stable

Stable growth typically includes:

- Real GDP remains around 1.5%–2.5%;
- Private final demand remains around 1.5%–2.5%;
- GDPNow is not repeatedly revised sharply upward or downward;
- Real consumption and business investment remain positive;
- GDP and GDI move in broadly the same direction.

This combination typically corresponds to a Goldilocks expansion.

Accelerating

Accelerating growth usually includes:

- GDP growth improves for two consecutive quarters;
- GDPNow is repeatedly revised upward;
- Consumption, business investment, and housing improve together;
- Private final demand exceeds 2%;
- PMI new orders and industrial production rise together.

If inflation also begins to rise, the economy is usually entering a reflationary phase.

Very Strong

Very strong growth usually includes:

- Real GDP above 3%;
- Private final demand above 3%;
- Real consumption and business investment both expand materially;
- GDPNow remains clearly above potential growth;
- Capacity utilization and new orders rise rapidly.

If employment is extremely tight, inflation is elevated, and the Federal Reserve continues tightening, the economy is usually overheating.

Positive but Slowing

This condition usually includes:

- GDP remains positive but falls toward 1%–2%;

- Private final demand slows;
- GDPNow is repeatedly revised modestly downward;
- Consumption remains positive, but durable goods and discretionary spending weaken;
- Business capital expenditure growth slows;
- Housing remains under pressure from high interest rates.

This combination typically corresponds to the late-cycle or soft-landing phase.

Slowing or Near Zero

This condition usually includes:

- Real GDP around 0%–1%;
- Private final demand near zero;
- GDPNow repeatedly revised downward;
- Real consumption growth becomes very weak;
- Business fixed investment stagnates;
- Manufacturing orders and industrial production decline.

If inflation remains elevated or reaccelerates, this usually corresponds to stagflation.

Negative Growth

This condition usually includes:

- GDP below 0%;
- Private final demand turns negative;
- GDP and GDI both decline;
- GDPNow becomes negative;
- Consumption, housing, and business investment contract together.

If layoffs rise and credit spreads widen at the same time, the economy is usually entering recession or a hard landing.

Deep Negative Growth

This condition usually includes:

- GDP below negative 2%;
- Private final demand contracts sharply;
- GDPNow is revised downward quickly;
- Business investment, consumption, and industrial production all fall broadly;
- Financial institutions and companies begin forced deleveraging.

This combination typically corresponds to a credit crisis.

Part III: Employment Indicators and Employment-State Assessment

1. Core Employment Indicators

The employment dimension should focus on:

- 1. Three-month average nonfarm payroll growth
- 2. Unemployment rate and the Sahm Rule
- 3. Initial unemployment claims
- 4. Continuing unemployment claims
- 5. JOLTS hiring rate
- 6. Job openings-to-unemployed ratio
- 7. Average weekly hours
- 8. Temporary-help employment

The three-month average of nonfarm payrolls is more stable than a single monthly reading.

The unemployment rate measures the share of people who are unemployed and actively seeking work.

Initial unemployment claims primarily reflect new layoffs.

Continuing claims reflect how difficult it is for unemployed workers to find new jobs.

The JOLTS hiring rate measures the pace at which companies are actually completing hires.

The job openings-to-unemployed ratio measures the degree of labor-market tightness.

2. Employment-State Thresholds

Employment State	Three-Month Average Payroll Growth	Unemployment Rate	Initial Claims	JOLTS Hiring Rate	Typical Cycle
Improving	Turns positive from negative, usually adding at least 50,000–150,000 jobs per month	Peaks or begins to decline	Falling repeatedly	Recovers from a low level	Early Recovery

3. Employment Indicators and Employment-State Assessment

Employment State	Three-Month Average Payroll Growth	Unemployment Rate	Initial Claims	JOLTS Hiring Rate	Typical Cycle
Healthy	Approximately 100,000–200,000 per month	Stable	Usually below roughly 220,000–240,000	Approximately 3.5%–4.0%	Goldilocks Expansion
Strong	Usually above 200,000 per month	Falling or remaining low	Remains low	Usually above 4%	Reflation
Extremely Tight	Usually above 200,000–250,000 per month	Clearly below long-run equilibrium	Very low	Clearly above long-run average	Overheating
Low Hiring, Low Layoffs	Approximately 50,000–150,000 per month and gradually slowing	Stable or rising modestly	Still below roughly 240,000	Below normal, often under 3.5%	Late Cycle / Soft Landing
Beginning to Deteriorate	Approximately 0–75,000 per month	Rises 0.2–0.4 percentage point from the low	Rises toward roughly 240,000–275,000	Continues to decline	Stagflation or Pre-Recession
Layoffs Increasing	Three-month average near zero or negative	Rising rapidly	Often above 275,000 or materially higher year over year	Very low	Recession / Hard Landing
Large-Scale Unemployment	Monthly payroll losses above 150,000	Rises rapidly; Sahm Rule clearly triggered	Usually above 350,000 and rising	Extremely low	Credit Crisis

3. Detailed Employment-State Combinations

Improving

Employment improvement usually includes:

- Payroll growth turns positive;
- Initial claims decline;
- Continuing claims peak;
- The JOLTS hiring rate begins to rise;
- Temporary-help employment stops falling;
- Average weekly hours stabilize.

This combination typically appears during an early recovery.

Healthy

A healthy labor market usually includes:

- Three-month average payroll growth of roughly 100,000–200,000 per month;
- Stable unemployment;
- Low initial claims;
- Continuing claims that are not rising materially;
- A JOLTS hiring rate within a normal range;
- Wage growth broadly consistent with productivity.

This combination typically corresponds to a Goldilocks expansion.

Strong

Strong employment usually includes:

- Three-month average payroll growth above 200,000;
- A declining unemployment rate;
- A high number of job openings;
- A rising hiring rate;
- Accelerating wage growth;
- Very low initial claims.

If inflation is also rising, this usually corresponds to reflation.

Extremely Tight

An extremely tight labor market usually includes:

- Unemployment clearly below long-run equilibrium;
- Job openings materially above the number of unemployed workers;
- Persistently high wage growth;

- Companies struggling to hire;
- High quit and hiring rates;
- Rising unit labor costs.

This combination usually corresponds to economic overheating.

Low Hiring, Low Layoffs

This condition usually includes:

- Companies reduce new job postings;
- The JOLTS hiring rate declines;
- Payrolls remain positive but weak;
- Initial claims remain low;
- Unemployment rises only modestly;
- Companies reduce hours rather than conduct large-scale layoffs.

This combination typically corresponds to late cycle or soft landing.

Beginning to Deteriorate

Employment begins to deteriorate when:

- Payroll growth approaches zero;
- Average weekly hours fall;
- Temporary-help employment declines;
- Continuing claims rise;
- Initial claims break above their previous stable range;
- Unemployment begins to accelerate upward.

If inflation remains high, this is more consistent with stagflation.

Layoffs Increasing

This condition usually includes:

- Three-month average payroll growth near zero or negative;
- Initial and continuing claims both rising;
- Corporate layoff announcements increasing;
- Unemployment rising materially;
- The Sahm Rule approaching or reaching a recessionary threshold.

This combination typically corresponds to recession or a hard landing.

Large-Scale Unemployment

This condition usually includes:

- Payroll losses exceeding 150,000 per month;
- Initial claims rising above 350,000;
- Unemployment increasing sharply over a short period;
- Continuing claims rising materially;
- Job openings and hiring rates both falling.

This combination typically corresponds to a credit crisis.

Part IV: Inflation Indicators and Inflation-State Assessment

1. Core Inflation Indicators

The inflation dimension should focus on:

1. **Core PCE year over year**
2. **Three- or six-month annualized core PCE**
3. **Three-month annualized core CPI**
4. **Core services inflation**
5. **Shelter inflation**
6. **Wage growth**
7. **Unit labor costs**
8. **Long-term inflation expectations**

Core PCE is the Federal Reserve's preferred measure of underlying inflation.

4. Inflation Indicators and Inflation-State Assessment

Core services and shelter inflation tend to be persistent.

Unit labor cost growth can be approximated as:

Wage growth minus productivity growth.

2. Inflation-State Thresholds

Inflation State	Core PCE, Year over Year	Three-Month Annualized Core CPI	Unit Labor Costs and Direction	Typical Cycle
Low	Usually below 2.5%	Usually below 2.5%	Low or declining	Early Recovery
Near 2%	Approximately 1.8%–2.5%	Approximately 1.5%–2.5%	Broadly consistent with a 2% inflation objective	Goldilocks Expansion
Rising	Increasing from previous levels	Usually above 3% and rising	Wage growth exceeds productivity	Reflation
High	Usually above 3%	Usually above 3.5%–4%	Persistently elevated	Overheating
Falling but Elevated	Year-over-year rate still around 2.5%–3.5%, but declining	Below the year-over-year rate and falling	Slowing	Late Cycle / Soft Landing
Rising or Elevated	Usually above 3% or reaccelerating	Above 3% and rising	Cost pressures increasing	Stagflation
Falling	Year-over-year rate steadily declining	Usually falls toward 1%–2.5%	Clearly declining	Recession / Hard Landing
Falling Rapidly	Rapidly approaching below 2%	Below 1% or negative	Falling sharply or turning negative	Credit Crisis

3. Detailed Inflation-State Combinations

Low

Low inflation usually includes:

- Core PCE below 2.5%;
- Three-month annualized core CPI below 2.5%;
- Slowing wage growth;
- Stable unit labor costs;
- Low goods and shelter inflation.

This combination is common in the late stages of recession and early recovery.

Near 2%

Inflation near target usually includes:

- Core PCE around 1.8%–2.5%;

- Three-month annualized core CPI around 1.5%–2.5%;
- No clear acceleration in services inflation;
- Stable long-term inflation expectations.

This combination typically corresponds to a Goldilocks expansion.

Rising

Rising inflation usually includes:

- Core PCE year over year reaccelerating;
- Three-month annualized core CPI above the year-over-year rate;
- Higher wage growth and unit labor costs;
- Inflation broadening across goods, shelter, or services.

If economic growth is also accelerating, this usually corresponds to reflation.

High

High inflation usually includes:

- Core PCE persistently above 3%;
- Three-month annualized core CPI above 3.5%–4%;
- Wage growth materially above productivity growth;
- Long-term inflation expectations beginning to rise;
- The Federal Reserve needing to continue tightening.

This combination usually corresponds to overheating.

Falling but Elevated

This condition usually includes:

- Core PCE still above 2.5%, but declining;
- Three-month annualized core CPI below the year-over-year rate;
- Shelter and services inflation gradually slowing;
- Unit labor cost growth declining;
- Long-term inflation expectations remaining stable.

This combination typically corresponds to late cycle or soft landing.

Rising or Elevated

This condition usually includes:

- Economic growth slowing while core inflation remains above 3%;
- Three-month annualized core CPI reaccelerating;
- Higher energy, tariff, or supply-chain costs;

- Corporate margins being squeezed by input costs;
- The Federal Reserve being unable to ease policy smoothly.

This combination typically corresponds to stagflation.

Falling

Falling inflation usually includes:

- Core PCE declining steadily;
- Three-month annualized core CPI falling toward 1%–2.5%;
- Wage and services inflation slowing;
- Companies losing pricing power.

If employment and credit are deteriorating at the same time, this usually corresponds to recession or a hard landing.

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2. Financial Conditions Indicators and State Assessment

This condition usually includes:

- Core inflation falling quickly toward below 2%;
- Three-month annualized core CPI below 1% or negative;
- Goods prices, wage growth, and asset prices all declining;
- Inflation risk shifting rapidly toward deflation risk.

This combination typically corresponds to a credit crisis.

Part V: Financial Conditions Indicators and State Assessment

1. Core Financial Conditions Indicators

Financial conditions should focus on:

1. **Federal funds target range**
2. **Real policy rate**
3. **10-year real Treasury yield**
4. **30-year mortgage rate**
5. **Financial conditions indexes**
6. **Bank lending standards**
7. **The U.S. dollar and long-term Treasury yields**
8. **The Federal Reserve balance sheet and quantitative tightening**

The federal funds target range reflects the Federal Reserve's policy stance.

The real policy rate can be approximated as the federal funds rate minus core inflation.

The 10-year real Treasury yield reflects long-term real financing costs and the discount rate applied to equity valuations.

2. Financial Conditions State Thresholds

Financial Conditions State	Policy Rate and Real Rate	10-Year Real Yield	Market and Banking Conditions	Typical Cycle
Accommodative	Policy rate below or near neutral; low real policy rate	Usually below 1%–1.5%	Bank lending improves; financial conditions index declines	Early Recovery
Neutral	Policy rate near neutral	Approximately 1%–1.5%	Lending standards stable	Goldilocks Expansion
Beginning to Tighten	Federal Reserve begins raising rates or postpones cuts	Rises by at least 0.5 percentage point	Mortgage and corporate financing costs rise	Reflation
Continuously Tightening	Real policy rate clearly positive	Usually above 1.5%–2% and rising	Quantitative tightening continues; lending standards tighten	Overheating
Restrictive	Real policy rate above roughly 1.5%	Usually around 1.5%–2.5% or higher	Mortgage, corporate, and bank credit conditions under pressure	Late Cycle / Soft Landing
Policy Dilemma	Growth weak, but rates cannot fall because inflation remains high	Remains elevated	Financial conditions remain tight	Stagflation
Beginning to Ease	Federal Reserve begins cumulative cuts of 25–100 basis points	Begins to decline	Market rates fall, but credit conditions may diverge	Recession / Hard Landing
Emergency Easing	Rapid cuts of 100–200 basis points or more	Falls sharply	Quantitative tightening stops and liquidity facilities are introduced	Credit Crisis

3. Detailed Financial Conditions Combinations

Accommodative

Accommodative conditions usually include:

- The Federal Reserve has already cut rates;
- The real policy rate is low;
- The 10-year real yield declines;
- Mortgage rates fall;
- Bank lending standards begin to ease;
- Financial conditions indexes decline.

This combination typically corresponds to an early recovery.

Neutral

Neutral conditions usually include:

- The policy rate near its long-run neutral level;
- Real interest rates neither especially high nor low;
- Stable bank lending standards;
- Normal corporate and housing financing;
- Limited volatility in the dollar and long-term yields.

This combination typically corresponds to a Goldilocks expansion.

Beginning to Tighten

The beginning of tightening usually includes:

- The Federal Reserve starts raising rates or delays cuts;
- The 10-year real yield rises;
- Mortgage and corporate bond yields increase;
- Financial conditions indexes tighten.

This combination typically corresponds to reflation.

Continuously Tightening

Continuous tightening usually includes:

- Repeated Federal Reserve rate increases;
- A materially higher real policy rate;
- Ongoing quantitative tightening;
- Tighter bank lending standards;
- Pressure on housing and high-valuation growth stocks.

This combination typically corresponds to overheating.

Restrictive

Restrictive conditions usually include:

- A clearly positive real policy rate;
- A 10-year real yield around 1.5%–2.5% or higher;
- High mortgage rates;
- Rising corporate refinancing costs;
- Tight bank lending standards.

If growth remains positive and employment does not deteriorate materially, this typically corresponds to late cycle or soft landing.

Policy Dilemma

A policy dilemma usually includes:

- Growth approaching stagnation;
- Unemployment beginning to rise;
- Core inflation remaining elevated;
- The Federal Reserve unable to cut rates rapidly;
- Real financing costs remaining high.

This combination typically corresponds to stagflation.

Beginning to Ease

The beginning of easing usually includes:

- The Federal Reserve starts cumulative rate cuts;
- Treasury yields decline;
- Real yields fall;
- Credit spreads may begin to widen.

If cuts occur while credit remains stable, the process may be consistent with a soft landing. If cuts occur alongside worsening employment and widening spreads, the process is more recessionary.

Emergency Easing

Emergency easing usually includes:

- Rapid Federal Reserve rate cuts;
- The end of quantitative tightening;
- Liquidity support for banks and markets;
- Treasury yields falling sharply;
- Private financing markets remaining frozen.

This combination typically corresponds to a credit crisis.

Part VI: Credit Indicators and Credit-State Assessment

1. Core Credit Indicators

The credit dimension should focus on:

- 1. U.S. high-yield option-adjusted spread
- 2. Investment-grade credit spreads
- 3. Speculative-grade corporate default rate
- 4. CCC-rated bond spreads
- 5. Bank C&I lending standards
- 6. Household delinquency rates
- 7. Credit-card and auto-loan delinquency rates
- 8. Bank nonperforming loan ratios
- 9. Corporate interest coverage ratios
- 10. Commercial real-estate loan quality

The high-yield OAS measures the additional compensation lower-rated companies must pay relative to Treasury securities.

The corporate default rate measures actual credit losses among highly leveraged firms.

Bank lending standards indicate whether banks are willing to provide credit to businesses.

Household delinquency rates reflect the condition of consumer balance sheets.

2. Credit-State Thresholds

Credit State	High-Yield OAS	Corporate Default Rate	Bank Lending and Household Credit	Typical Cycle
Repairing	Declines from above 500 basis points toward below 400 basis points	Peaks and declines	Degree of tightening eases; delinquencies stabilize	Early Recovery
e. Credit Indicators and Credit-State Assessment				
	basis points	2%-4%	delinquencies stable	expansion

Credit State	High-Yield OAS	Corporate Default Rate	Bank Lending and Household Credit	Typical Cycle
Still Loose	Usually below 300–350 basis points	Usually below 3%	Banks remain willing to lend	Reflation
High Leverage	Spreads remain low, but corporate and household leverage rises	Temporarily low	Debt growth and asset prices rise	Overheating
Tail-End Stress	Overall OAS remains below 400–450 basis points, but CCC spreads are clearly wider	Defaults rise among low-rated companies	Credit-card, auto-loan, and lower-income delinquencies rise	Late Cycle / Soft Landing
Increasing Stress	Approximately 450–600 basis points	Approximately 4%–6%	Banks tighten materially; delinquencies accelerate	Stagflation
Spreads Widening	Approximately 600–800 basis points	Usually above 6%	Investment-grade spreads widen; bank losses increase	Recession / Hard Landing
Credit Freeze	Usually above 800–1,000 basis points	Usually above 8%–10%	Banks stop lending; bond issuance freezes	Credit Crisis

3. Detailed Credit-State Combinations

Repairing

Credit repair usually includes:

- High-yield OAS declining from elevated levels;
- Corporate default rates peaking;
- Bank lending standards no longer tightening further;
- Bond issuance recovering;
- Household delinquencies stabilizing.

This combination typically corresponds to an early recovery.

Stable

Stable credit usually includes:

- High-yield OAS around 300–450 basis points;
- Corporate default rates around 2%–4%;

- Stable investment-grade spreads;
- Bank lending standards near neutral;
- Household delinquencies not rising materially.

This combination typically corresponds to a Goldilocks expansion.

Still Loose

Loose credit usually includes:

- High-yield OAS below 300–350 basis points;
- Low corporate default rates;
- Banks willing to lend;
- Smooth bond issuance;
- Strong market risk appetite.

If growth and inflation are also rising, this usually corresponds to reflation.

High Leverage

High leverage usually includes:

- Credit spreads remaining low;
- Corporate and household debt rising;
- More leveraged acquisitions and lower-quality issuance;
- Rising asset prices;
- Borrowers becoming overly confident about a low-default environment.

This state typically appears during overheating.

Tail-End Stress

Tail-end stress usually includes:

- Overall high-yield OAS remaining relatively low;
- CCC spreads widening materially;
- Credit-card and auto-loan delinquencies rising;
- Problems appearing in commercial real estate and small-business lending;
- Stress not yet spreading to investment-grade companies or large banks.

This combination typically corresponds to late cycle.

Increasing Stress

Increasing credit stress usually includes:

- High-yield OAS rising toward 450–600 basis points;
- Default rates rising toward 4%–6%;

- Banks tightening lending standards materially;
- Household delinquencies accelerating;
- Corporate interest coverage ratios declining.

If growth is weak but inflation remains high, this usually corresponds to stagflation.

Spreads Widening

Widening credit spreads usually include:

- High-yield OAS rising toward 600–800 basis points;
- Investment-grade spreads also widening;
- Corporate default rates above 6%;
- Higher bank loan losses;

▲ Five-Dimension Indicators and Economic Cycle Mapping

This combination typically corresponds to recession or a hard landing.

Credit Freeze

A credit freeze usually includes:

- High-yield OAS above 800–1,000 basis points;
- Corporate default rates above 8%–10%;
- Banks stopping or sharply reducing lending;
- Companies unable to issue debt normally;
- Investors being forced to sell assets;
- Financial institutions experiencing capital or liquidity stress.

This combination typically corresponds to a credit crisis.

Part VII: Five-Dimension Indicators and Economic Cycle Mapping

Cycle Assessment	Growth	Employment	Inflation	Financial Conditions	Credit
Early Recovery	GDP turns positive, private demand improves, GDPNow revised upward	Payrolls turn positive, claims decline, hiring rate improves	Core inflation low or still declining	Rate cuts already delivered; real yields falling	OAS narrows; default rate peaks

Cycle Assessment	Growth	Employment	Inflation	Financial Conditions	Credit
Goldilocks Expansion	GDP and private demand around 1.5%–2.5%	Healthy payroll growth; unemployment stable	Core inflation near 2%	Policy rate near neutral	OAS and delinquency rates stable
Reflation	GDP and private demand accelerate toward 2.5%–3% or more	Hiring and wages reaccelerate	Core PCE and short-term CPI momentum rise	Cuts delayed; real yields rise	Spreads remain low; financing loose
Overheating	GDP and private demand usually above 3%	Unemployment extremely low; labor market very tight	Core inflation above 3%; broad cost pressure	Federal Reserve continues tightening	Spreads low, but leverage accumulates
Late Cycle / Soft Landing	GDP remains positive but slows toward 1%–2%	Low hiring, low layoffs	Inflation falls but remains above target	Real rates remain restrictive	Overall stable, but lower-quality borrowers under pressure
Stagflation	GDP near zero; private demand weak	Hiring declines; unemployment rises gradually	Inflation elevated or reaccelerating	Federal Reserve unable to ease	OAS and delinquencies begin to rise
Recession / Hard Landing	GDP, GDI, and private demand turn negative	Payrolls turn negative; claims and unemployment rise	Inflation falls materially	Federal Reserve begins rapid easing	OAS widens; defaults and losses rise
Credit Crisis	GDP contracts deeply	Large-scale layoffs and unemployment	Inflation falls rapidly or turns into deflation	Emergency easing and liquidity support	Financing freezes; banks stop lending

Part VIII: Economic Cycle and Equity-Market Performance

Cycle Assessment	S&P 500	Nasdaq	Rate-Sensitive Equities	Defensive Equities	Growth Equities
Early Recovery	Usually performs very strongly as the market begins pricing in an earnings rebound	Usually strong, benefiting from rate cuts and recovering risk appetite	Usually very strong, especially real estate, homebuilders, regional banks, and small caps	Usually underperforms the broader market	Usually strong, especially high-valuation growth stocks that fell sharply earlier
Goldilocks Expansion	Rises steadily as earnings growth broadens	Advances steadily, but does not always materially outperform the S&P 500	Performs reasonably well, depending on interest-rate stability	Usually underperforms	High-quality growth companies with deliverable earnings perform well
Reflation	Usually rises, with cyclicals, industrials, energy, and financials leading	May rise, but often begins to lag the S&P 500	Mixed performance: banks may benefit, while REITs and utilities may be pressured by higher rates	Usually underperforms	High-valuation growth stocks are vulnerable to rising long-term yields
Overheating	May continue rising initially, but volatility increases later	Often comes under pressure, especially long-duration and highly valued technology stocks	Most rate-sensitive groups weaken, although banks may benefit temporarily from higher rates	Begins to gain a relative advantage	Most vulnerable to valuation compression

8: Economic Cycle and Equity-Market Performance

Cycle Assessment	S&P 500	Nasdaq	Rate-Sensitive Equities	Defensive Equities	Growth Equities
Late Cycle / Soft Landing	The index may continue rising, but sector dispersion widens	May regain leadership if inflation falls and rates peak; remains volatile if long-term yields stay high	Real estate, housing, and regional banks may remain weak; utilities depend on rate-cut expectations	Relative performance improves	Profitable, cash-generative large growth companies outperform loss-making growth companies
Stagflation	Usually performs poorly because both margins and valuations are pressured	Usually weak due to the combination of high rates and low growth	Most groups perform poorly, especially REITs, housing, and utilities; banks may also be hurt by credit losses	Relatively resilient, especially consumer staples and healthcare	Usually weak, particularly high-valuation companies dependent on distant earnings
Recession / Hard Landing	Falls materially in the early phase, but may bottom early as rate cuts are priced in	Falls sharply initially; may lead the rebound once inflation declines and easing is confirmed	Falls initially because of weaker earnings and credit, but may rebound early after rate cuts	Usually relatively resilient	Weak initially, but high-quality growth often leads the subsequent recovery

Cycle Assessment	S&P 500	Nasdaq	Rate-Sensitive Equities	Defensive Equities	Growth Equities
Credit Crisis	Usually falls sharply, with financials and cyclicals among the weakest sectors	Falls sharply initially; high-quality cash-generative technology is relatively resilient but still vulnerable	Banks, real estate, homebuilders, and leveraged companies are usually among the worst performers	Relatively resilient, although absolute returns may still be negative	Unprofitable and highly leveraged growth companies are weakest; cash-rich large technology companies perform relatively better

Part IX: Typical Sequence of the Economic Cycle

1. Full Main Sequence

The most common full economic-cycle sequence is:

Recession or Credit Crisis → Late Recession or Economic Bottom → Early Recovery →

3. Typical Sequence of the Economic Cycle

After the late-cycle stage, three major paths emerge:

Late-Cycle Slowdown → Soft Landing, Stagflation, or Hard Landing

2. Path After a Soft Landing

The typical soft-landing sequence is:

Late-Cycle Slowdown → Inflation Continues Falling → Federal Reserve Cuts Gradually → Real Rates Decline → Consumption and Investment Stabilize → Goldilocks Expansion

A soft landing does not require the economy to enter a formal recession.

The economy can complete the disinflation process while growth remains low but positive, employment remains stable, and the credit system continues functioning.

Equity markets usually price this outcome before the economy visibly reaccelerates. High-quality growth stocks, the S&P 500, and some rate-sensitive sectors may rise before the economic data improves materially.

3. Path After Stagflation

Stagflation can lead to two different outcomes.

The first path is:

Stagflation → Energy and Supply Shocks Ease → Inflation Falls → Federal Reserve Begins Cutting Rates → Soft Landing or Moderate Recovery

The second path is:

Stagflation → Inflation Remains Elevated → Federal Reserve Maintains High Rates or Tightens Further → Employment and Credit Deteriorate → Hard Landing or Recession

Whether stagflation ends smoothly depends largely on whether inflation is driven by temporary supply shocks or has spread into wages, services, and long-term inflation expectations.

4. Path After a Hard Landing

The typical hard-landing sequence is:

Late-Cycle Slowdown → Layoffs Increase → Consumption Falls → Corporate Profits Decline → Credit Spreads Widen → Recession → Federal Reserve Cuts Rapidly → Economic Bottom → Early Recovery

If the credit system remains functional, the hard landing may remain a conventional recession.

If corporate defaults, bank losses, and financing freezes reinforce one another, the downturn may evolve into a credit crisis.

5. Path After a Credit Crisis

The typical post-credit-crisis sequence is:

Credit Freeze → Corporate and Household Deleveraging → Emergency Federal Reserve Easing → Government Financial and Fiscal Support → Credit Spreads Peak → Economic Bottom → Early Recovery

Recovery after a credit crisis usually takes longer because companies, households, and financial institutions must repair their balance sheets.

Part X: Causes of the Three Late-Cycle Pathways

1. Causes of a Soft Landing

The core condition for a soft landing is:

Inflation continues to decline while employment and credit remain stable.

A soft landing is usually supported by the following factors.

Companies Reduce Hiring but Avoid Large-Scale Layoffs

Companies control costs by:

- Reducing job openings;
- Slowing new hiring;
- Cutting overtime and average hours;
- Slowing wage growth;
- Delaying some expansion plans.

As long as companies avoid mass layoffs, household income and consumption can remain positive.

Inflation Falls Through Gradual Demand Cooling

As consumption and hiring slow:

- Wage growth decelerates;
- Rent inflation falls;
- Service-price growth slows;
- Corporate pricing power weakens;
- Core PCE gradually approaches target.

Productivity Improves

If corporate investment improves productivity, wages can continue rising without causing excessive unit labor cost growth.

AI, automation, software, equipment, and infrastructure investment may all support higher productivity.

The Credit System Remains Intact

A soft landing generally requires:

- Stable high-yield OAS;

10. Causes of the Three Late-Cycle Pathways

- Household delinquency stress not spreading beyond lower-income groups;
- Normal access to investment-grade financing.

Federal Reserve Timing Is Appropriate

The Federal Reserve must avoid two mistakes:

- Cutting too early and allowing inflation to reaccelerate;
- Cutting too late and allowing employment and credit to deteriorate sharply.

Fiscal Policy Remains Moderate

The government can support growth through infrastructure, housing, energy, and productivity investment, but should avoid large, broad-based consumption stimulus while inflation remains elevated.

2. Causes of Stagflation

The core stagflation combination is:

Growth slows while inflation remains high or reaccelerates.

Stagflation is usually caused by the following factors.

Energy-Price Shocks

Higher oil, natural-gas, and electricity prices raise:

- Household living costs;
- Corporate transportation costs;
- Industrial production costs;
- Airline and logistics expenses.

Higher energy prices reduce household real purchasing power while simultaneously raising inflation.

Tariffs and Trade Frictions

Tariffs increase the cost of imported goods and intermediate inputs.

Companies may pass these additional costs to consumers, increasing inflation. At the same time, lower margins and weaker investment can slow growth.

Supply-Chain Disruptions

Wars, natural disasters, port problems, or trade restrictions can cause:

- Raw-material shortages;

- Higher shipping costs;
- Longer delivery times;
- Insufficient inventories.

These supply shocks create weak growth and high inflation at the same time.

Lower Labor Supply

If labor-force participation declines or labor-force growth is inadequate, companies may face wage pressure even as the economy slows. Services inflation may therefore remain elevated.

Lower Productivity

When wage growth exceeds productivity growth, unit labor costs rise. Companies must either raise prices or accept lower profit margins.

Conflict Between Fiscal Stimulus and Monetary Tightening

If the government continues supporting demand through large deficits while the Federal Reserve keeps rates high to control inflation, the two policies work against each other.

Fiscal policy supports growth and demand, while monetary policy suppresses demand and financing. Together they may produce:

- Persistent inflation;
- Elevated long-term yields;
- Crowding out of the private sector;
- Gradually weaker economic growth.

Federal Reserve Policy Dilemma

If the Federal Reserve cuts rates, inflation may reaccelerate. If it keeps rates high, housing, business investment, and employment may weaken further.

3. Causes of a Hard Landing

The core hard-landing combination is:

Inflation begins to fall, but employment and credit deteriorate rapidly.

A hard landing is usually caused by the following factors.

Interest Rates Remain High for Too Long

Monetary policy operates with a lag.

High rates gradually affect:

- Corporate refinancing;
- Mortgages;
- Commercial real estate;
- Auto loans;
- Credit cards;
- Small-business loans;
- Venture capital and growth-company financing.

Even after the Federal Reserve stops raising rates, the earlier tightening continues to affect the economy.

Corporate Profits Decline

When revenue growth slows while wage, interest, and depreciation costs remain elevated, profit margins decline.

Companies may then cut:

- Hiring;
- Wage growth;
- Capital expenditure;
- Inventories;
- Research and development;
- Marketing spending.

Capital Expenditure Falls Sharply

If earlier investment in AI, data centers, equipment, real estate, or infrastructure produces weaker-than-expected returns, companies may reduce capital expenditure rapidly.

Lower capital spending affects:

- Equipment suppliers;
- Semiconductor and industrial companies;
- Construction firms;
- Software companies;
- Power and infrastructure businesses;
- Local labor markets.

Consumer Buffers Are Exhausted

Households may simultaneously face:

- Depleted excess savings;
- Rising credit-card balances;
- High interest rates;
- Slowing real-income growth;

- Higher unemployment risk;
- Falling asset prices.

Once employment weakens, consumers may cut discretionary spending rapidly.

A Credit Event Occurs

Problems in commercial real estate, high-yield debt, private credit, or regional banks may trigger tighter credit conditions.

When banks and investors become concerned about losses, they raise financing standards, making it difficult even for previously healthy companies to borrow.

Fiscal Policy Tightens Abruptly

If the government cuts spending, raises taxes, or reduces transfers while private demand is already weak, the economy receives an additional negative shock.

The Federal Reserve Cuts Too Late

If the Federal Reserve waits until unemployment and credit risks are already rising quickly, rate cuts may not stop the recession immediately because companies and households have already begun deleveraging.

II. Comparison of the Three Late-Cycle Paths

Part XI: Comparison of the Three Late-Cycle Paths

Factor	Soft Landing	Stagflation	Hard Landing
Growth	Slows but remains positive	Near zero or low growth	Turns negative
Employment	Low hiring but low layoffs	Weak hiring; unemployment rises gradually	Layoffs rise rapidly
Inflation	Continues falling	Remains elevated or reaccelerates	Falls rapidly
Federal Reserve	Cuts gradually	Unable to ease smoothly	Cuts rapidly
Fiscal Policy	Provides moderate support	Demand stimulus may worsen inflation	Usually requires stronger support
Credit	Broadly stable	Deteriorates locally	Deteriorates broadly
Corporate Profits	Grow modestly or decline slightly	Revenue slows and margins compress	Decline materially

Factor	Soft Landing	Stagflation	Hard Landing
Equities	S&P 500 and quality growth relatively strong	Defensive and energy sectors relatively stronger	Defensives lead early; growth and cyclicals rebound later
Next Stage	Returns to Goldilocks expansion	Moves toward either soft landing or hard landing	Recession followed by an economic bottom and early recovery

Part XII: Final Decision Rules

The direction after the late-cycle stage is mainly determined by two questions:

- Can inflation continue to decline?
- Can employment and credit remain stable?

The corresponding outcomes are:

- **Falling inflation with stable employment and credit: Soft Landing.**
- **Persistent inflation with continued growth deterioration: Stagflation.**
- **Falling inflation with rapidly deteriorating employment and credit: Hard Landing.**
- **Reaccelerating growth and inflation: Reflation or No Landing.**
- **Deep contraction with banking and credit-market dysfunction: Credit Crisis.**

The most common full-cycle sequence is:

Recession or Credit Crisis → Economic Bottom → Early Recovery → Goldilocks Expansion → Reflation → Overheating → Late-Cycle Slowdown → Soft Landing, Stagflation, or Hard Landing.

15. Final Decision Rules